A Leakage-Aware Benchmark for Ultra-Small Language Models on Structured Speech-Command Routing

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Abstract. *Small language models are attractive for local command interfaces, but a valid-looking response can still be semantically wrong or impossible to execute. We present a reproducible benchmark of five instruction-tuned checkpoints with at most three billion parameters on a transcript-to-command task. The benchmark derives a narrow two-operation contract from Fluent Speech Commands (FSC), a human-recorded English speech-command corpus: four native semantic combinations map to `media_control` or `volume_adjust`, while all other examples receive an explicitly derived policy-abstention target. We evaluate Qwen2.5-0.5B, Qwen3.5-0.8B, TinyLlama-1.1B, SmolLM2-1.7B, and Qwen3.5-2B with the same canonical prompt and greedy decoding on an NVIDIA RTX 5090. Two balanced protocols separate official speaker disjointness from normalized-template disjointness, and all reports include item-level and template-cluster statistics. On the phrase-disjoint test, a transparent lexical control reaches 77.40% exact match, whereas the best model reaches 50.00% exact match; its 100% contract-valid rate is explained by near-universal abstention rather than successful call routing. Qwen3.5-0.8B reaches 43.60% exact match and 60.15% contract validity, while the other three models obtain zero exact matches under the strict contract. The result is a benchmark and leakage audit, not evidence of Android control, automatic speech recognition, or on-device performance. It shows that structured-output compliance and conservative policy behavior must be measured separately from semantic accuracy in the ultra-small regime.*

Keywords: ultra-small language models; structured prediction; speech commands; function routing; abstention; data leakage; reproducible benchmarking.

1. Scope and Research Questions

This article deliberately narrows the original mobile-assistant project to a benchmark of ultra-small language models. The parameter cap is strict: no checkpoint with more than 3,000,000,000 parameters is included. The input is an English human speech transcription, and the output is a validated JSON object. The benchmark does not claim Brazilian-Portuguese coverage, Android execution, automatic speech recognition (ASR) quality, battery behavior, thermal behavior, or smartphone latency. The phone is not required.

We ask: (i) how do checkpoints below the parameter cap differ in JSON compliance and canonical command accuracy under one declared contract? (ii) how much does exact-match performance change when normalized transcription templates are held out, rather than only speakers? (iii) do simple controls provide a meaningful reference for the models? and (iv) how different are item-level and template-cluster summaries when repeated phrasings occur across speakers?

2. Human Corpus and Derived Task

We use Fluent Speech Commands (FSC), introduced as a corpus for end-to-end spoken-language understanding [1]. It contains 30,043 human recordings, 97 speakers, and 31 native intents, with official train, validation, and test partitions. We use the transcriptions and split metadata, not the waveforms. The raw archive remains outside Git and is identified by SHA-256 in the server-local manifest.

FSC is not a command-routing or Android corpus. We therefore define an explicit, generic two-operation contract. Only the following four native combinations become calls:

Table 1. The four deterministic FSC-to-contract bridge rules.

Native action	Native object	Derived target
activate	music	`media_control(action=play)`
deactivate	music	`media_control(action=pause)`
increase	volume	`volume_adjust(direction=up)`
decrease	volume	`volume_adjust(direction=down)`

Every other retained example receives `{action: abstain, tool: null, arguments: {}}`. This is an experimenter-defined policy label, not a human FSC annotation. To prevent the larger set of unsupported intents from dominating the score, the generator samples a balanced 1:1 mixture of calls and policy abstentions within each split. Each derived dataset contains 14,956 records: 7,478 calls and 7,478 abstentions.

The contract checks exactly three top-level fields (`action`, `tool`, and `arguments`), restricts tool names to the two declared operations, and validates the required argument enums. It is an evaluation target only; it does not execute a tool or grant any permission.

3. Leakage-Audited Protocols

The first protocol preserves FSC's official speaker-disjoint split. It measures transfer to held-out speakers but does not prevent repeated normalized transcriptions from crossing partitions. The second protocol assigns each Unicode-normalized, case-folded, whitespace-normalized and punctuation-stripped transcription template to one split within its native FSC label. Speakers may occur in more than one split in this second protocol by design.

Table 2. Balanced derived datasets and leakage controls.

Protocol	Train	Dev	Test	Total	Control
Official speaker-disjoint	11,442	1,580	1,934	14,956	speakers
Phrase-disjoint	10,458	2,202	2,296	14,956	templates

The official protocol has zero speaker intersections but 245, 247, and 244 template intersections for train–dev, train–test, and dev–test, respectively. The phrase-disjoint protocol has zero template intersections. It contains only 38 test template clusters, so cluster intervals are intentionally wide; this is a limitation to report, not a reason to pool clusters with individual examples.

4. Models and Execution

The matrix contains five checkpoints. Parameter counts are computed by summing tensor shapes in the local safetensors files; this makes the cap auditable and includes all checkpoint components. Qwen3.5 checkpoints are exposed by Transformers as multimodal conditional-generation models, but no image or video input is supplied in this transcript-only experiment.

Table 3. Checkpoints included in the strict ultra-small matrix.

Checkpoint	Parameters	Family	Revision
Qwen2.5-0.5B-Instruct	494,032,768	Qwen2	7ae5576
Qwen3.5-0.8B	873,438,784	Qwen3.5	2fc0636
TinyLlama-1.1B-Chat	1,100,048,384	Llama	fe8a4ea
SmolLM2-1.7B-Instruct	1,711,376,384	SmolLM	31b70e2
Qwen3.5-2B	2,274,069,824	Qwen3.5	15852e8

The excluded Qwen2.5-3B-Instruct checkpoint has 3,085,938,688 measured parameters and therefore exceeds the strict cap. Qwen3.5-4B-Base is also excluded. Complete IDs, revisions, licenses, and counts are recorded in `benchmarks/ultrasmall_models.json`.

All models use their native chat template with the same English semantic instruction, compact catalog, greedy decoding (`do_sample=false`), and a 64-token generation cap. The template mode and model class are recorded per prediction. Runs use Transformers 5.3.0, PyTorch 2.10.0+cu128, and one NVIDIA GeForce RTX 5090 with 32 GB VRAM. No fine-tuning is part of the main matrix. A previously completed Qwen2.5-0.5B LoRA run is retained as an exploratory artifact and is not used in the model comparison.

5. Metrics and Controls

The evaluator distinguishes parseability and JSON validity; contract validity, including action/tool/argument constraints; exact match against the derived target; action accuracy; tool and argument accuracy on call gold items; and abstention precision, recall, and F1.

The primary accuracy comparison is item-level exact match accompanied by template-level summaries. For each `template_id`, the evaluator computes mean exact-match rate, mean action accuracy, mean abstention F1, and the strict proportion of clusters whose items are all exact. Cluster-bootstrap 95% intervals resample template IDs, not rows. Wilson intervals are retained as descriptive item-level intervals but are not treated as leakage-aware uncertainty.

Two deterministic controls use only the transcription text and contract. `always_abstain` returns policy abstention for every item. `lexical` uses transparent regular expressions over media/volume and play/pause/increase/decrease cues; it never reads FSC native labels or metadata.

6. Results

6.1. Main Phrase-Disjoint Comparison

The lexical control outperforms every zero-shot model on exact match. Qwen3.5-2B is contract-valid on every item but obtains only 50.00% exact match: its output behavior

Table 4. Phrase-disjoint test results. Percentages are item-level except for the final column.

System	Parse	Valid	Exact	Abstain F1	Template exact
Always abstain	100.00	100.00	50.00	66.67	76.32
Lexical control	100.00	100.00	**77.40**	81.56	**89.47**
Qwen2.5-0.5B	70.17	0.00	0.00	–	0.00
Qwen3.5-0.8B	100.00	60.15	43.60	69.92	50.00
TinyLlama-1.1B	43.42	0.00	0.00	–	0.00
SmolLM2-1.7B	96.47	0.00	0.00	–	0.00
Qwen3.5-2B	100.00	100.00	50.00	66.67	76.32

is dominated by abstention, which is correct for policy negatives but wrong for calls. Qwen3.5-0.8B is the only model with a substantial mixture of valid calls and abstentions, yet its exact match remains below the lexical control. The other three checkpoints produce parseable text often enough to be measurable, but their responses do not satisfy the declared canonical schema.

6.2. Official Speaker-Disjoint Comparison

Table 5. Official speaker-disjoint test results.

System	Valid	Exact	Template exact
Always abstain	100.00	50.00	75.71
Lexical control	100.00	**77.92**	**89.07**
Qwen2.5-0.5B	0.00	0.00	0.00
Qwen3.5-0.8B	73.47	54.65	56.28
TinyLlama-1.1B	0.00	0.00	0.00
SmolLM2-1.7B	0.78	0.00	0.00
Qwen3.5-2B	100.00	50.98	76.11

The official split gives Qwen3.5-0.8B a higher exact-match rate than phrase-disjoint (54.65% versus 43.60%), an observed split gap of 11.05 percentage points when template overlap is removed. We treat this divergence as an observational comparison rather than a formal causal test: the template-cluster bootstrap 95% CI for phrase-disjoint exact match is wide ([26.02%, 63.35%]), reflecting the sample of 38 held-out template clusters (9 call-only and 29 abstain-only clusters, due to the diversity of unsupported FSC intents mapped to abstention). Qwen3.5-2B changes from 50.98% to 50.00%, while the lexical control is comparatively stable (77.92% to 77.40%). The official split therefore cannot be interpreted as evidence of lexical generalization.

6.3. Structural Error Audit and Latency

The failure modes are qualitatively different. Qwen2.5-0.5B frequently emits JSON-like objects with an operation name in `action`, rather than the required `action=call` plus a tool field. TinyLlama frequently explains the answer or reaches the 64-token cap before closing the object. SmolLM2 often emits a plausible JSON object with the wrong field structure. These cases explain why parseability is not interchangeable with contract validity.

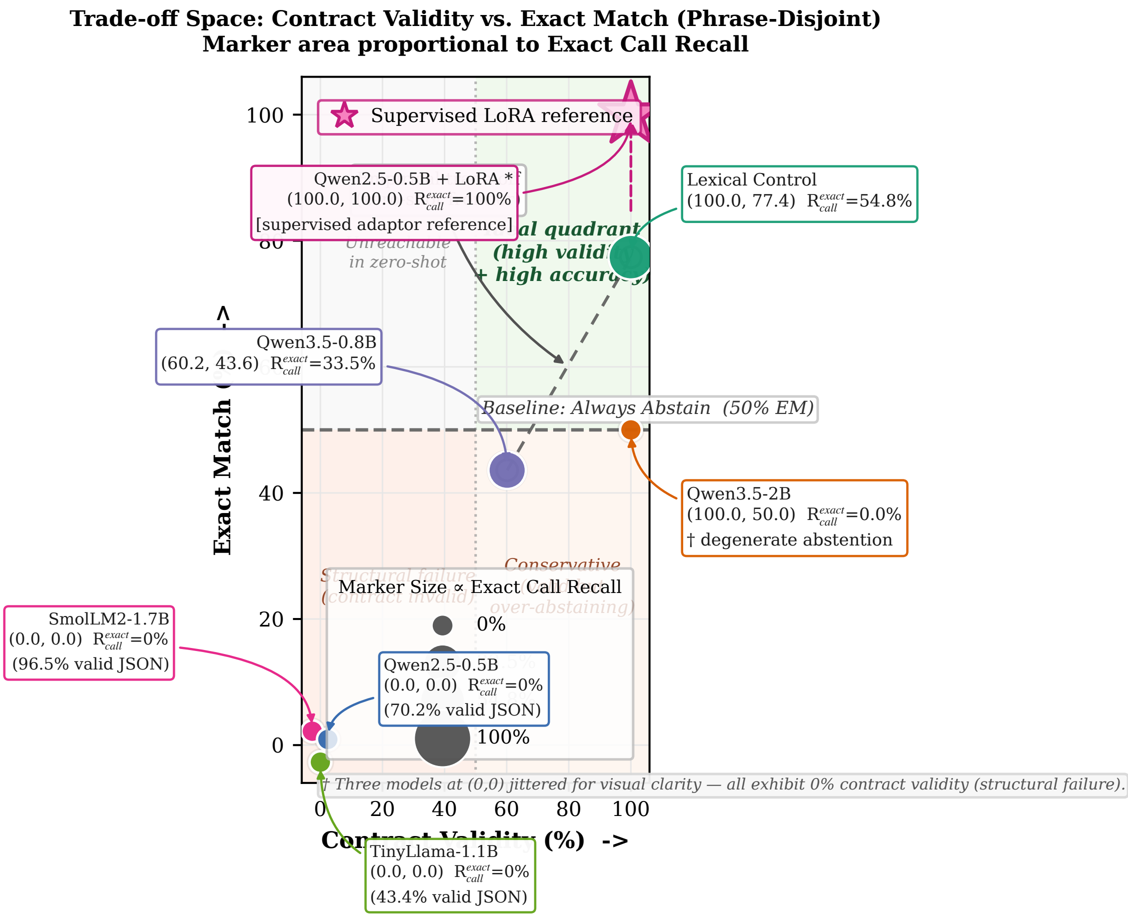

Figure 1. Trade-off space on the phrase-disjoint test ($n = 2{,}296$ items, 38 template clusters). x: contract validity (%), y: exact match (item-level %). Marker area $\propto$ exact call recall (proportion of gold calls with correct tool and arguments). Qwen3.5-2B attains 100% validity via degenerate abstention (exact call recall 0%); lexical control (100.0, 77.4) dominates the zero-shot systems; Qwen3.5-0.8B (60.2, 43.6) trades validity for calls. Three models overlap at (0,0) with 0% validity and are jittered for legibility (values in parentheses are JSON-valid rates). Star: Qwen2.5-0.5B+LoRA adaptor (100, 100), non-zero-shot supervised reference.

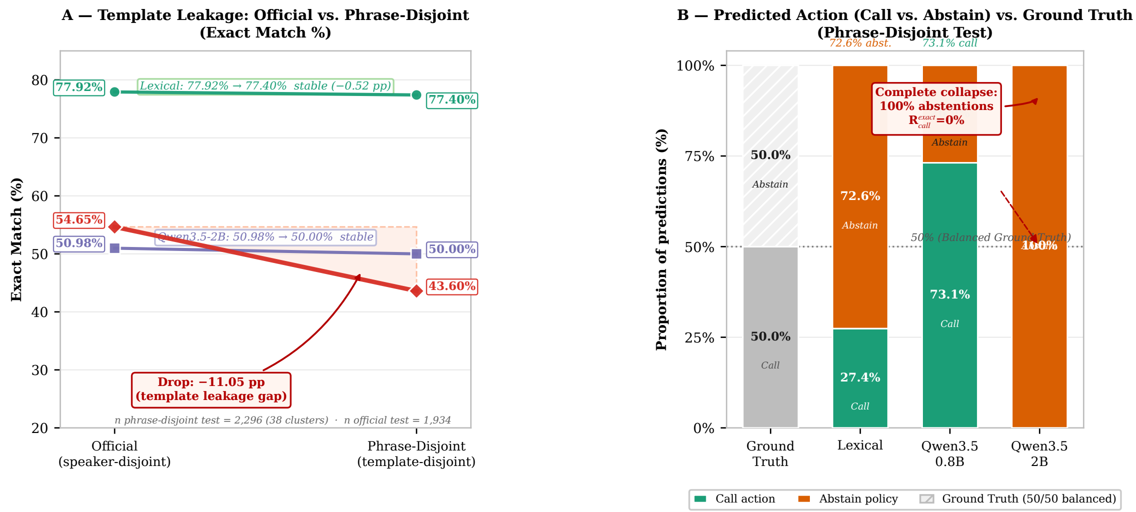

Figure 2. Template leakage and policy collapse (phrase-disjoint test is 50% call / 50% abstain). A Exact match on official (speaker-disjoint, $n = 1{,}934$) vs. phrase-disjoint ($n = 2{,}296$, 38 template clusters). Lexical: 77.92% $\rightarrow$ 77.40% (-0.52 pp, stable); Qwen3.5-2B: 50.98% $\rightarrow$ 50.00% (stable, degenerate abstention); Qwen3.5-0.8B: 54.65% $\rightarrow$ 43.60% (-11.05 pp, shaded gap) with template-cluster 95% CI [26.02, 63.35] on phrase-disjoint. B Predicted action distribution vs. balanced ground truth. Lexical: 27.4% call / 72.6% abstain; Qwen3.5-0.8B: 73.1% call / 26.9% abstain; Qwen3.5-2B: 0% call / 100% abstain (complete collapse, exact call recall 0).

Server-side single-request latency is reported only as an engineering observation. On phrase-disjoint, mean / median / p95 milliseconds were: Qwen2.5-0.5B 81.4 / 91.0 / 176.0; Qwen3.5-0.8B 324.1 / 327.2 / 448.9; TinyLlama 301.5 / 300.2 / 311.6; SmolLM2 122.9 / 106.7 / 220.1; and Qwen3.5-2B 345.1 / 338.4 / 395.9. These values depend on server load, Transformers implementation, prompt length, and decoding behavior; they are not smartphone measurements.

7. Discussion

The benchmark does not support a monotonic “more parameters is better” conclusion. The 2.274B Qwen3.5 checkpoint is structurally compliant but conservative to the point of abstaining on calls. The 0.873B Qwen3.5 checkpoint routes more calls but loses both contract validity and exactness on unseen templates. The external-family checkpoints demonstrate that instruction tuning and model size alone do not guarantee compatibility with a strict output contract.

The lexical control is essential. Without it, the 50% balanced abstention baseline could make a model that refuses everything appear competitive. Cluster analysis is also essential: repeated templates give some systems much larger effective row counts than the number of independent phrasings. The phrase-disjoint test has only 38 clusters, so its uncertainty is broad; nevertheless, it answers a different question from speaker transfer and should not be merged with the official result.

The derived abstention labels require particular caution. They express the declared benchmark policy, not human judgments that a command is unsafe or unsupported in a production assistant. A deployed system would need an independent policy layer, schema

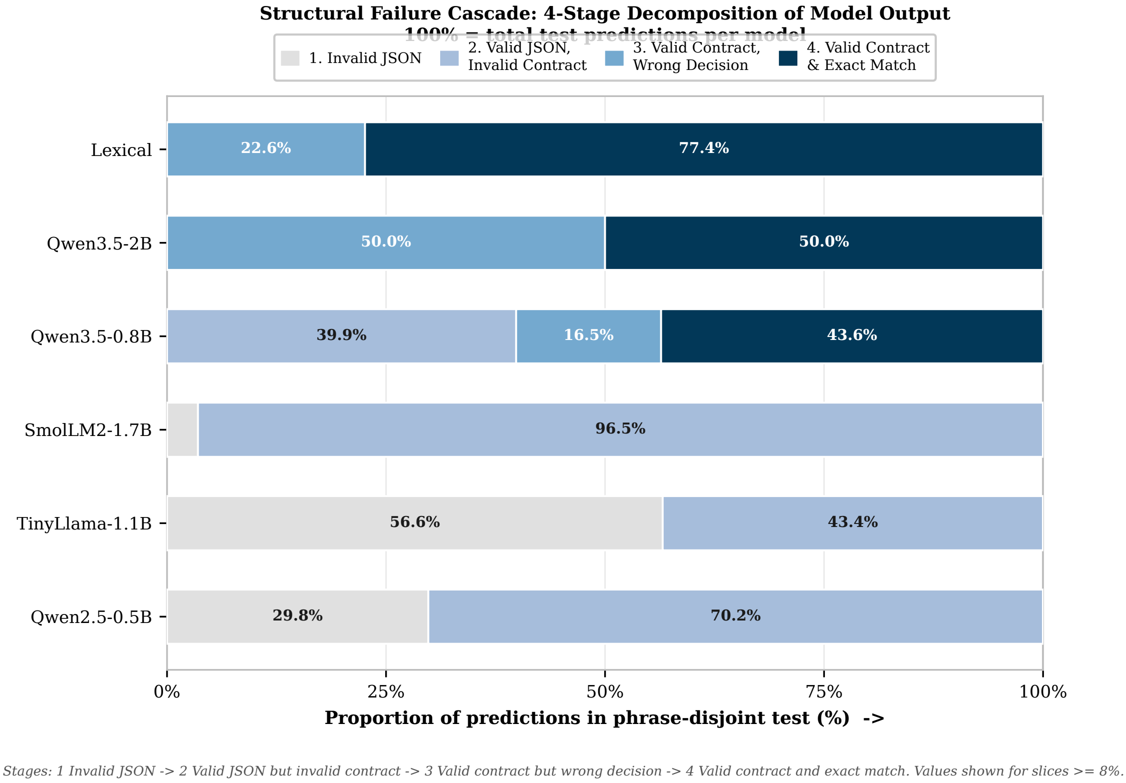

Figure 3. Structural cascade on the phrase-disjoint test ($n = 2{,}296$). Each bar is 100% of a system's predictions decomposed into four mutually exclusive stages: (1) invalid JSON, (2) valid JSON but contract-invalid, (3) contract-valid but wrong decision, (4) contract-valid and exact. Lexical and Qwen3.5-2B have no structural failure (stages 1–2 $= 0$); Qwen3.5-0.8B loses 39.85 pp at stage 2, while Qwen2.5-0.5B, TinyLlama-1.1B and SmolLM2-1.7B obtain 0% exact matches because all parseable output falls in stage 2. Labels shown only for slices $\geq 8\%$.

validator, confidence handling, and execution authorization. The present experiment stops before all of those layers.

In contrast, two techniques would change the reading of Figures 1–3 but are deliberately kept outside the zero-shot matrix. The retained Qwen2.5-0.5B+LoRA adaptor [2] (star in Figure 1) reaches 100% contract validity and 100% exact match on the same phrase-disjoint split; as a supervised in-distribution adaptor it shows the contract is learnable, not that zero-shot routing is solved, and is therefore not commensurable with the greedy zero-shot results. Constrained decoding via grammar-enforced generation (GBNF grammars in `llama.cpp`, Outlines, Guidance) would likewise guarantee 100% validity by construction and collapse stages 1–2 of Figure 3 to zero. This enforces syntactic compliance without implying semantic correctness: a grammar can force a valid `{action, tool, arguments}` shape while leaving call-vs-abstain discrimination and argument exactness to the model. We therefore keep the main benchmark unconstrained and report validity separately from exact match, treating LoRA and grammar-constrained decoding as explicit, non-zero-shot baselines for future work.

8. Reproducibility and Data Governance

The Neuromancer repository records the model matrix, checkpoint revisions, parameter-count method, generic contract, deterministic FSC generator, both split manifests, validator, controls, common inference script, cluster-aware evaluator, runner, logs, and aggregate reports. From the repository root, the reproducible main command (offline, reusing cached predictions) is:

```
HF_HUB_OFFLINE=1 python scripts/run_ultrasmall_benchmark.py \
  --skip-training --skip-existing
```

If an isolated interpreter is required, prefix with the local environment, e.g., `.venv/bin/python scripts/run_ultrasmall_benchmark.py`.

The raw FSC archive and derived JSONL files remain server-local. The source license is recorded as CC BY-NC-ND 4.0 for academic research; no audio, transcript, or per-example prediction is redistributed in Git. Aggregate metrics are sufficient to audit the reported claims without publishing restricted data.

9. Limitations and Next Gates

1. FSC is an English smart-home/assistant corpus, not a native Android or Brazilian-Portuguese command corpus.
2. The four call labels and all abstention labels are derived by a manually declared bridge.
3. The task consumes transcriptions, so ASR errors and acoustic variability are not evaluated.
4. The phrase-disjoint test has only 38 unique template clusters.
5. Native chat templates differ across model families; this is logged but remains a possible prompt-format confound.
6. The main matrix is zero-shot and uses one deterministic decoding setting and one checkpoint revision per model; it does not estimate training variance.
7. No model output is executed, and no result is measured on the phone.

10. Conclusion

We provide a leakage-aware, reproducible benchmark for five ultra-small language models on structured routing from human-origin speech transcriptions. The central result is negative but useful: under a strict canonical contract, a transparent lexical control outperforms every zero-shot model, while the largest model's perfect contract validity is mostly universal abstention. The benchmark therefore separates output compliance, policy abstention, and semantic exactness, and makes phrase leakage visible. It is a defensible first article for the ultra-small-model benchmark scope; Android deployment, ASR, Portuguese human data, and physical-device evaluation are future studies rather than claims of this paper.

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